

Response to Reviewers — Round 2

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Title: OphDDI-CDS: An Absorption-First Deterministic Cascade for Ophthalmic Drug–Drug Interaction Screening

Authors: Wang Guoqing, M.D.; Yi Xianglong, M.D., Ph.D. (corresponding)

Dear Dr Shelke, Dr Parikesit and reviewers,

We thank both reviewers for an unusually careful read. Reviewer 1 in particular audited the deposit rather than trusting it, and several of the findings are correct in ways we could not have discovered from our own working copy: the deposit did not contain the evaluation material, two of the three stubs really were stubs, and the absorption forcing does not operate where the manuscript said it does. We verified each point independently before acting on it, and where a claim was wrong we have said so in the manuscript rather than quietly correcting it.

This round we have completed the changes that are purely textual or documentary. Three items remain open, and we state clearly which they are and why, rather than letting the response imply more progress than the manuscript reflects.

A marked-changes copy is uploaded as a related file, with every changed paragraph in red. The main manuscript file is clean, as the journal requires.

Reviewer 1

Comment 1 — the deposit did not support the reproducibility claims made for it

Accepted; the material is now deposited. The reviewer is right that a fresh clone contained no gold file, no expert ratings, no prediction rows, no evidence base and no DDInter severity table, and that configs/prompts.yaml and

src/ophthalmic_ddi_cds_agent/naive_rag.py were placeholders. The material existed in the predecessor working copy; it had never been deposited, and the Availability sentence described a package the reader could not reach. That was a genuine failure of the deposit, not a misunderstanding.

All of it is now public at <https://github.com/wangguoqing932/ophddi-cds> (release tag v1.0-submission, accessible without an account), and the Availability statement has been rewritten to describe what is actually there.

Requested	Now deposited
Four gold files with case-level provenance	data/gold/ — 6 sets (118 / 92 / 200 / 70 + two development-gate sets), each with cases.jsonl, dataset.meta.json and source references
9,600 prediction rows	data/predictions/ — 4 methods × 5 temperature conditions × 4 datasets, per-seed files, aggregate metrics, run log
12-case expert ratings	data/expert_review/ — ratings, response archive, review instrument, scanned questionnaires
Evidence base	data/evidence/evidence_chunks.jsonl — 1,307 chunks, plus the offline citation cache
DDInter severity table	data/ddinter/ddinter_code_{A,B,C,N,S}.csv
Exact prompts	configs/prompts.yaml — was 1 line, now 98: system prompt, BM25 parameters, per-method user prompts, parse rule, 10-token cap, temperature schedule
Retrieval code	src/ophthalmic_ddi_cds_agent/naive_rag.py — was a 2-line skeleton, now a

Requested

Now deposited

working BM25 retrieval class

Evaluation and statistics scripts

scripts/run_multiseed_per_dataset.py, scripts/evaluate.py

scripts/run_baselines.py retains its --synthetic behaviour, but its module docstring now states in its first lines that it is an offline pipeline scaffold, **not** the reported evaluation, and that its synthetic mode hardcodes full_system at zero error and must never be used as a result.

Comment 2 — “40 citation-verified mechanism rules” is not what the report shows

Accepted in full. The verification report covers 8 manifest items over 7 distinct DOIs, of which three rules resolve to a verified literature citation. Of the 40 rules, 53 source records break down as 41 DrugBank entries, 7 regulatory label records, 2 DOIs, 2 DDInter pair annotations and 1 PubMed citation; the DrugBank and label records were never independently re-verified.

The phrase “citation-verified” has been removed throughout, and Methods Section 2.2 now states the coverage precisely, including the reviewer’s suggested formulation. We have also recorded the two specific problems the reviewer identified: that several rules whose output class would call for a citation declare a list that does not resolve to a verified item, and that beta_blocker_digoxin_additive cites a study of additive topical-plus-systemic beta-blockade rather than a digoxin-specific source while grading as high a pair its own DDInter annotation records as moderate. The conservative high grade is retained; the provenance gap is now stated rather than left implicit.

Comment 3 — the absorption-first mechanism is not enforced where the paper says it is

Accepted; the reviewer’s numbers reproduce exactly. We wrote an independent audit (scripts/path_attribution_audit.py, output deposited as audits/path_attribution.json) and swept all 26,216 evaluable registry pairs. The result matches the reviewer’s figures: **171 pairs with a low-or-below absorption tier receive medium (95) or high (76), all 171 arrive**

through the rule path, and all 76 high grades come from nsaid_antihypertensive_antagonism, which requires no absorption flag.

The manuscript no longer claims that absorption scaling forces low-absorption pairs to low risk across the cascade. Section 2.2 now states that the forcing applies within the matrix layer only and that the final grade is the maximum of an ungated rule layer and an absorption-scaled matrix layer, and Section 4.2 draws the consequence. The path distribution the reviewer asked for is reported: class matrix 68.8%, G19 hard constraint 12.0%, rule layer 11.9%, default-low 7.3%. The previously undocumented default-low path is now described in the cascade description.

Open: whether to gate nsaid_antihypertensive_antagonism on absorption, as the four matrix upgrades are gated. That changes the system's behaviour and therefore the reported results, so we have not done it silently; we describe the behaviour accurately in this revision and identify the gating decision as future work.

Comment 4 — the absorption tier is the most load-bearing and least documented variable

Accepted; the documentation gaps are closed, the reassignment is flagged.

- **Per-agent table.** Now deposited as Supplementary Table S5 (113 rows) with a source record for every agent, and in the deposit at `data/registry/per_agent_absorption_tier.csv`. The reviewer is right that the previous Supplementary Table S5 was the audit stratification and that the word flurbiprofen did not appear anywhere in the supplement.
- **Operational definition.** Section 2.2 now gives the thresholds and units: the tier is a categorical assignment intended to represent fractional systemic bioavailability (high $\geq 50\%$, medium $\geq 10\text{--}50\%$, low $<10\%$, with very-low, minimal and none indicating progressively smaller exposure), and it is stated explicitly that the sources report heterogeneous quantities (bioavailability fractions, plasma concentrations, qualitative statements), so the tiers are an ordinal model rather than measured exposure.

- **Six values, not four.** Section 2.2 now reports all six and their counts (16 high, 17 medium, 60 low, 9 very-low, 7 minimal, 4 none) and notes that the L3 constraint tests four of them.
- **flurbiprofen.** Confirmed: tiered high in entities_a.csv while the candidate file records no tier flag, with identical evidence sources. Recorded as a defect in Section 4.4.
- **Nine non-topical agents.** Confirmed: six intravitreal anti-VEGF agents (including aflibercept, which the manuscript itself uses as the example of a pair with no plausible topical pathway), intraocular acetylcholine and two surgical viscoelastics. Recorded in Section 4.4.

Open: re-running the registry analysis after correcting the assignments. Because the tier drives both the class-matrix forcing and the L3 constraint, correcting it will move the 75.0% figure in Section 3.4 and other reported numbers. We have therefore not changed the assignments in this revision; instead we deposit the tier table with sources, report the defects, and make the correction the first item of future work (Section 4.5). Section 3.4 already carries the tier cross-tabulation the reviewer asked for (audits/divergence_by_tier.csv), so the magnitude can be judged against the tier distribution that produces it.

Comment 5 — the baseline comparison cannot support the claim made for it

Accepted; the conclusion is qualified rather than defended. Section 2.4 now states that the comparison used a single proprietary endpoint (deepseek-chat) with a 10-token output cap that forces a one-word answer and precludes any reasoning or justification, that parsing was by substring, that the named lightrag baseline does not query the LightRAG index, and that these choices bound what the comparison can support. The exact prompts and parameters are provided in Supplementary Text S3 and in configs/prompts.yaml.

We have also reported the comparison symmetrically, as requested: on the audit set the two ungrounded baselines reach 0.848 high-risk sensitivity each, against 0.440 for the

cascade (lightrag 0.296). This appears in Section 3.5 and, in abbreviated form, in the Abstract.

Open: re-running the comparison with a reasoning model, adequate token budget and repeated sampling at fixed temperature. This is a substantive new experiment rather than a revision of text, and we would prefer to run it as such; the current revision states what the existing comparison can and cannot support.

Comment 6 — the registry-scale divergence measures the absorption model, not the database

Partially accepted. The framing point is taken: Section 3.4 already disclosed that the tiers were assigned by the study team, and it now leads with the consequence — the measurement is a property of an explicitly stated absorption model, not an independent pharmacokinetic result. The tier $\times$ severity cross-tabulation the reviewer asked for is deposited as audits/divergence_by_tier.csv. We have also softened the Background claim about route-specific severity being absent from mainstream resources, and added the reviewer's examples (Drugs.com maintains a separate ophthalmic entity for timolol with its own severity distribution; Lexicomp asks the user to specify route).

Open: the same analysis under one or two alternative tier assignments. This depends on the tier reassignment in comment 4 and is folded into the same future-work item.

Comment 7 — the expert argument is a tautology as stated

Accepted; the sentence has been removed. The reviewer's logic is right: because the system reproduced its gold on all 12 cases, expert-versus-gold and expert-versus-system agreement are necessarily the same comparison, and that identity cannot be evidence that the divergence is not system-specific.

In the course of acting on this we found a second, related problem that neither reviewer could have seen from the manuscript. **The expert instrument was not blinded.** The questionnaire shown to both ophthalmologists displays, for every case, the proposed gold grade and the system grade with a one-line mechanistic rationale; the scanned questionnaires, the blank template and the archived responses all carry these two columns.

The manuscript previously described the experts as “blinded to system outputs and gold labels”, which is not correct. We have rewritten the passage as a **grade review** in Methods 2.5 (renamed), Results 3.7, the contributions paragraph, Discussion 4.1, Limitations 3.8 and 4.4, and the Abstract and Conclusion.

What the exercise does support is stated plainly and nothing more: two ophthalmologists, given no mechanism evidence, did not endorse the assigned grades, and their divergence was stable between them; with $n = 12$ and near-chance kappa it cannot be distinguished from chance. We have also noted, as the reviewer requested, that the maximum achievable unweighted kappa of 0.625 applies to one expert’s marginals rather than both.

Comment 8 — the audit set mixes a third-party component with one graded under our assumption

Accepted. Section 3.5 now reports 37 of 62 (**0.597**) alongside 0.707 and identifies it as the figure free of the system’s own assumption; the Abstract carries the same qualification. The stratified breakdown was already disclosed and remains.

Comment 9 — supplementary cross-references are systematically wrong

Accepted; the numbering has been rebuilt. The root cause was that prose sections and tables shared one S1–S6 sequence while the tables also carried their own Table S1–S6 labels, so the two collided. There are now two independent series, **Supplementary Text S1–S7** and **Supplementary Table S1–S8**, and every cross-reference in the manuscript has been checked against the item it names.

The four missing items the reviewer identified have been supplied:

- the **per-agent absorption tier table** (now Table S5, 113 rows with sources);
- the **evaluation prompts** (now Text S3, full system prompt and all three user prompts with parameters);
- the **TRIPOD-AI checklist** (now Text S6 — the previous Text S6 was the abbreviation list, which is now Text S7);
- the **L1 disagreement rationales** (now Table S1 — the previous Table S1 was the registry summary, now Table S4).

Comment 11 — matrix counts

Accepted. The configuration declares 21 ophthalmic row keys and 40 systemic column keys, 34 columns in use, and 104 cells carrying an explicit level. Section 2.2 now reports these figures and states that the earlier $20 \times 37 / 82$ description did not match the configuration. The malformed cell (anticholinergic $\times$ antidepressant_tca, whose advice string had been split by a newline-and-colon into a spurious delirium): null key) has been repaired, and the matrix is described as sparse by design with a default cell for undefined pairs.

Comment 12 — “zero variance” as a comparative virtue

Accepted. Removed from the Abstract and the Conclusion. It is retained in Results 3.6, where it is stated as a determinism check on a method that makes no model call, and in Section 4.2 it is explicitly recast as a design property rather than a robustness result.

Comment 13 — test suite requires forced UTF-8

Partially confirmed. The skipped test is confirmed to be an empty placeholder (test_integration.py, marked “Fill in during Phase 4”). We have not been able to reproduce the cp1252 failure on a clean Windows clone: the repository contains no Greek beta in any file the tests read, and the suite reports 34 passed, 1 skipped under the default console encoding. We would be grateful for the traceback so we can fix the specific case rather than guessing.

Comment 14 — fingerprints depend on line endings

Accepted; the definition is now stated. Section 2.7 notes that SHA-256 digests are defined over **LF-normalised content**, gives both digests for the deposited configs/rules.yaml (2afabaad81161c40 under CRLF as deposited, b5c77f6fde8a9f7d after LF normalisation) so a reader can reproduce either, and states that the evidence and index fingerprints are recorded alongside the deposited artefacts rather than only in the project log.

Comment 15 — reference errors

Accepted; all four corrected, verified against Crossref and the arXiv API:

Ref	Was	Now
1	fifth author “Peterson G”	Widmark G
14	Cochrane issue 3	issue 2
21	authors “Wang Y, Wu Z, Wu C, et al.”	Guo Z, Xia L, Yu Y, Ao T, Huang C (5 authors; the previous list named people not on the paper)
27	2019, no volume	2020;48(1):24-30

The statement “three over-alerts” has been corrected to **four**: tacrolimus × ibuprofen was omitted from the list.

Reviewer 2

All fourteen points have been addressed in Methods Sections 2.1–2.4.

#	Point	Where addressed
1	Whether route of administration is covered	New paragraph in 2.1: route is modelled through the absorption tier and through route-dependent rule predicates, not as a free-text field
2	Bold L1/L2/L3 and other sub-labels	L1/L2/L3 now bold in 2.2; the corresponding labels in 3.5 bolded
3	Second-order interactions	2.2: none are modelled;

#	Point	Where addressed
		rules are evaluated independently; the only cross-layer composition is the final maximum
4	Most matrix cells undefined	2.2: the matrix is sparse by design, 104 defined cells, undefined pairs fall through to the L3 default rather than receiving an interpolated grade; the matrix is deposited
5	Is absorption scaling empirically validated	2.2: stated that it is not, with the rationale and where its behavioural consequence is quantified
6	Which layer the class mapping applies to	2.2: the drug-to-class mapping is consumed by the L2 matrix layer only
7	Gold selection criteria	2.3: inclusion rule for the 20 mechanism annotations and the sampling logic for the 58 guideline cases
8	“No human expert participated” vs a study team of experts	2.3: clarified that the team comprises the two named clinician authors, so the statement means no <i>independent</i> expert was involved
9	How the gold shares	2.3: the sharing differs by

#	Point	Where addressed
	sources with the rules	subset, described per subset
10	Contradictory evidence	2.3: more conservative grade retained, disagreement recorded in the case file
11	Development data in its own subsection; correction method	2.3: development sets separated from validation sets, features and labels described; explicit three-condition correction rule for the 30 audit cases
12	DeepSeek architecture version and input language	2.4: model identifier as used, the endpoint does not expose architecture, English input throughout
13	RAG protocol	2.4: BM25 parameters, query string, top-k, truncation and prompt assembly given in full
14	Prompts not findable	Supplementary Text S3 (full prompts) and <code>configs/prompts.yaml</code> in the deposit

Open items, stated plainly

We prefer the response to be shorter than the record of what remains undone.

1. **Absorption-tier reassignment (R1-4, R1-6).** The defects are documented and the per-agent table is deposited, but the tiers themselves have not been changed,

because doing so moves the 75.0% divergence figure and other reported numbers. We have made this the first item of future work rather than presenting corrected numbers we have not produced.

2. **Rule-layer absorption gating (R1-3).** The cascade's actual behaviour is now described accurately, with the path distribution quantified. Whether to gate nsaid_antihypertensive_antagonism is a design decision with downstream effects on results, and we have left it open.
3. **Baseline re-run with a reasoning model (R1-5).** Not performed. The comparison's limitations are now stated explicitly in both the Methods and the Discussion, and we have reported the direction in which the baselines outperform the system.

We are grateful to Reviewer 1 for auditing the deposit rather than accepting it, and to Reviewer 2 for a set of questions that made the Methods substantially more complete.

Yours sincerely,

Wang Guoqing, M.D. Department of Ophthalmology, The First Affiliated Hospital of Xinjiang Medical University E-mail: 17690924120@163.com

Yi Xianglong, M.D., Ph.D. (corresponding author) Professor of Ophthalmology, Department of Ophthalmology, The First Affiliated Hospital of Xinjiang Medical University E-mail: yixianglong1010@163.com